REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 9 Advanced | Module 1 Lesson 4 | Graphing Data

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Slope = $\Delta y/\Delta x$; linear: $y = mx + b$; quadratic: $y = kx^2$; inverse: $y = k/x$.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 17–22.

P01. Which variable is deliberately changed by the investigator?

A. Independent variable B. Dependent variable
C. Controlled variable D. Random error

Answer

A. Independent variable.

P02. The dependent variable is usually plotted on the:

A. Horizontal axis B. Vertical axis
C. Page border D. Graph title only

Answer

B. Vertical axis.

P03. A straight line through the origin is evidence of:

A. An inverse relationship B. A constant nonzero intercept
C. Direct proportionality D. A quadratic relationship

Answer

C. Direct proportionality.

P04. For y proportional to x^2, doubling x makes y:

A. Twice as large B. Half as large
C. Unchanged D. Four times as large

Answer

D. Four times as large.

P05. A best-fit line should:

A. Represent the overall trend in the data B. Join every point by sharp segments
C. Always pass through the origin D. Avoid all measured points

Answer

A. Represent the overall trend in the data.

P06. A straight graph of force F (N) against extension x (cm) passes through (1, 2) and (5, 14). Calculate its slope.

A. 3 N/cm B. 6 N/cm
C. 1.5 N/cm D. -3 N/cm

Answer

A. 3 N/cm. Slope = $\Delta F/\Delta x = (14 - 2)/(5 - 1) = 3$ N/cm.

P07. A measured relationship is y = 4x + 3, where x is in seconds and y is in metres. Find y when x = 4 s.

A. 38 m B. 19 m
C. 9.5 m D. -19 m

Answer

B. 19 m. $y = 4 \times 4 + 3 = 19$ m.

P08. A data model is $y = 2x^2$, with x in seconds and y in metres. Calculate y at x = 3 s.

A. 36 m B. 9 m
C. 18 m D. -18 m

Answer

C. 18 m. $y = 2 \times 3^2 = 18$ m.

P09. A data model is y = 48/x. Find y when x = 6. Both x and y are dimensionless.

A. 16 B. 4
C. -8 D. 8

Answer

D. 8. $y = 48/6 = 8$.

P10. Three repeated measurements are 12, 14 and 16 s. Calculate their mean.

A. 14 s B. 28 s
C. 7 s D. -14 s

Answer

A. 14 s. Mean = (12 + 14 + 16)/3 = 14 s. Averaging can reduce random variation.

P11. Define a controlled variable.

Answer

A factor kept the same so that the effect of the independent variable can be investigated fairly.

P12. What does the slope of a graph represent?

Answer

The change in the vertical quantity divided by the corresponding change in the horizontal quantity, with units determined by the axes.

P13. What does a nonzero y-intercept mean?

Answer

The fitted relationship predicts a nonzero y value when x = 0; its physical meaning depends on the experiment.

P14. How can a graph indicate an inverse relationship?

Answer

As x increases, y decreases in the form y = k/x; the product xy is constant for ideal data.

P15. Why should axis labels include units?

Answer

They identify the quantities and scales so readers can interpret values and calculate meaningful slopes.

P16. A straight graph of force F (N) against extension x (cm) passes through (1, 2) and (7, 20). Calculate its slope.

Answer

Slope = $\Delta F/\Delta x = (20 - 2)/(7 - 1) = 3$ N/cm.

P17. A straight graph of force F (N) against extension x (cm) passes through (1, 2) and (9, 26). Calculate its slope.

Answer

Slope = $\Delta F/\Delta x = (26 - 2)/(9 - 1) = 3$ N/cm.

P18. A measured relationship is y = 6x + 3, where x is in seconds and y is in metres. Find y when x = 4 s.

Answer

$y = 6 \times 4 + 3 = 27$ m.

P19. A measured relationship is y = 8x + 3, where x is in seconds and y is in metres. Find y when x = 4 s.

Answer

$y = 8 \times 4 + 3 = 35$ m.

P20. A data model is $y = 2x^2$, with x in seconds and y in metres. Calculate y at x = 4 s.

Answer

$y = 2 \times 4^2 = 32$ m.

P21. A data model is $y = 2x^2$, with x in seconds and y in metres. Calculate y at x = 5 s.

Answer

$y = 2 \times 5^2 = 50$ m.

P22. A data model is y = 72/x. Find y when x = 9. Both x and y are dimensionless.

Answer

$y = 72/9 = 8$.

P23. A data model is y = 96/x. Find y when x = 12. Both x and y are dimensionless.

Answer

$y = 96/12 = 8$.

P24. Three repeated measurements are 13, 15 and 17 s. Calculate their mean.

Answer

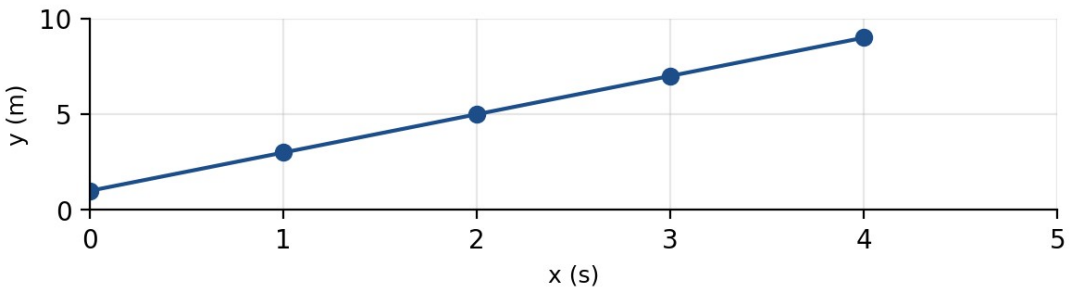
Mean = (13 + 15 + 17)/3 = 15 s. Averaging can reduce random variation.

P25. Three repeated measurements are 14, 16 and 18 s. Calculate their mean.

Answer

Mean = (14 + 16 + 18)/3 = 16 s. Averaging can reduce random variation.

P26. Plot x = 0, 1, 2, 3, 4 s against y = 1, 3, 5, 7, 9 m. Draw a best-fit line and state its slope and intercept.



Answer

The graph is a straight line y = 2x + 1. Slope = 2 m/s; y-intercept = 1 m. It is linear but not directly proportional because it does not pass through the origin.

P27. A student calls the data x = 1, 2, 3 and y = 3, 6, 9 an inverse relationship. Evaluate the claim.

Answer

The claim is false. y/x = 3 is constant, so y is directly proportional to x. The products xy are not constant.

P28. Describe a graphing procedure for noisy experimental data.

Answer

Choose suitable labelled axes and scales, plot measured points accurately, draw a line or curve that represents the overall trend and use well-separated points on the best-fit line for slope.

P29. Data follow $y = 4x^2$. Suggest a transformation that makes a straight-line graph.

Answer

Plot y against x^2. The model becomes $y = 4(x^2)$, a straight line through the origin with slope 4 in the appropriate transformed units.

P30. An apparent straight line does not pass through the origin although theory predicts proportionality. Suggest two checks.

Answer

Check for zero offsets in instruments and whether the variables were correctly defined or background readings subtracted. Consider measurement uncertainty before forcing the fit through the origin.